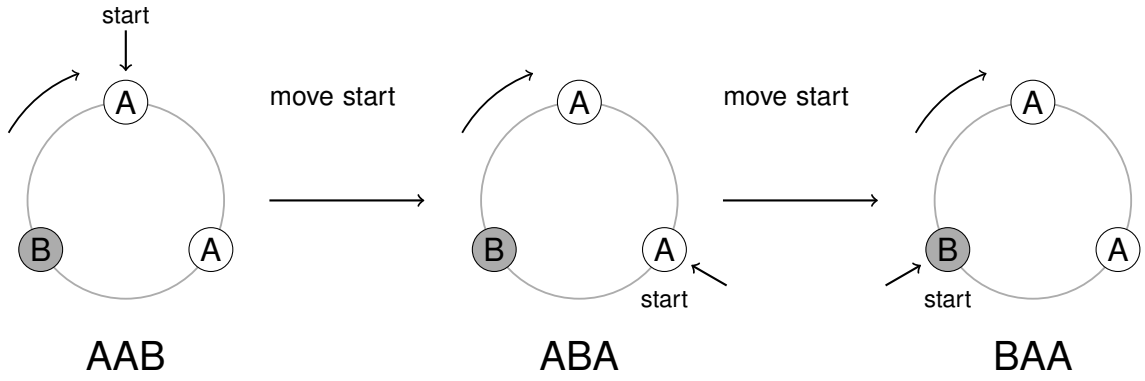
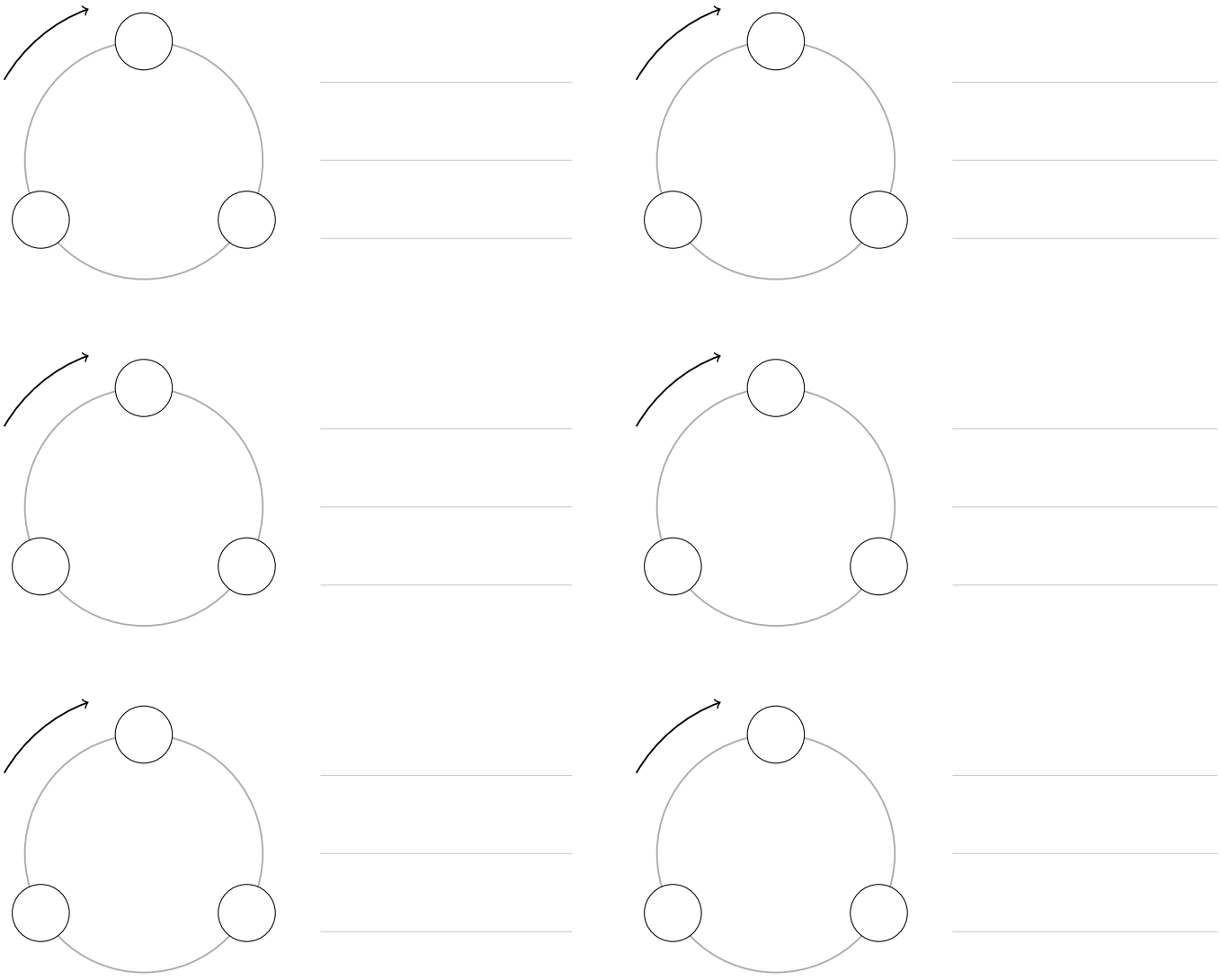
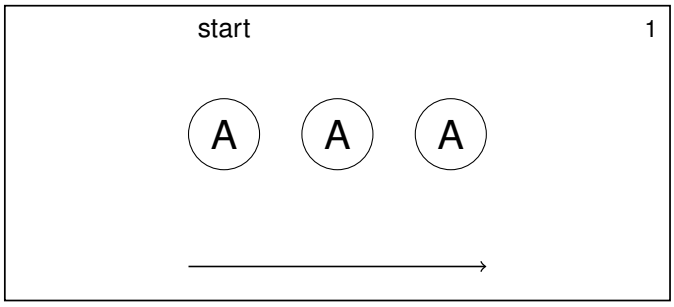
Each bead may be A or B; one-color rings are allowed. Read clockwise from the start arrow. The bead order is the readout. Rings count as the same when turning makes them match. Keep them face up; do not flip them over.

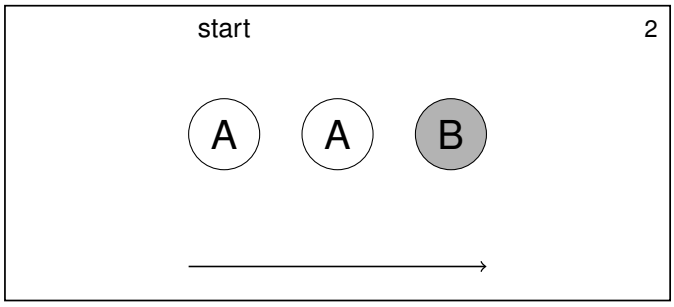


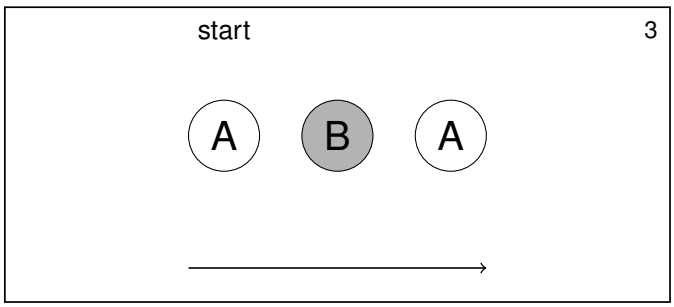
Problem 1: Make every different three-bead ring using A and B. Find all its readouts as you move the start.

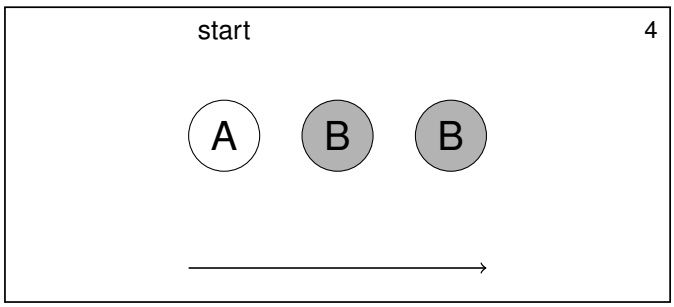


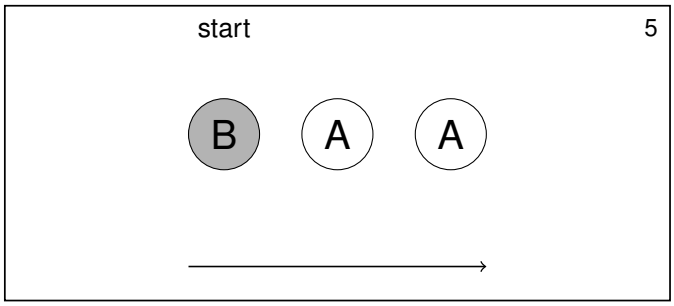
Problem 2: Group cards that make the same ring. How many groups are there, and how many cards are in each group?

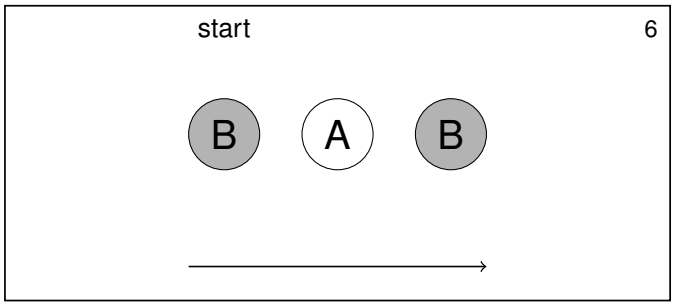


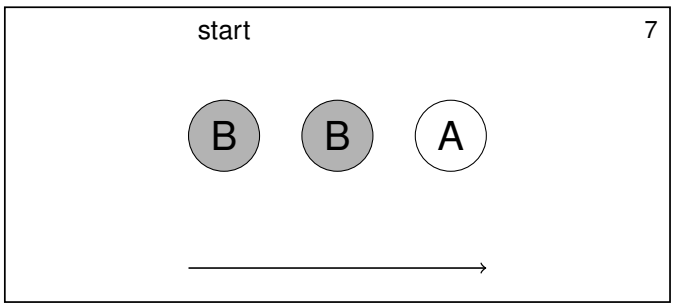


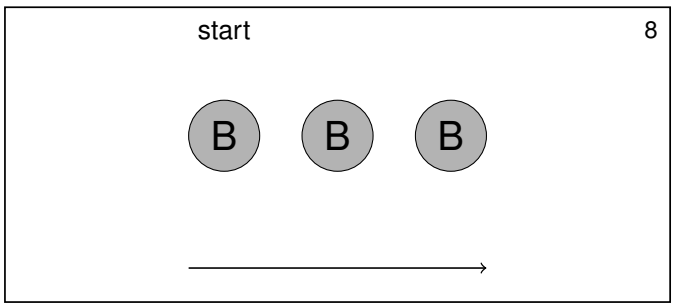




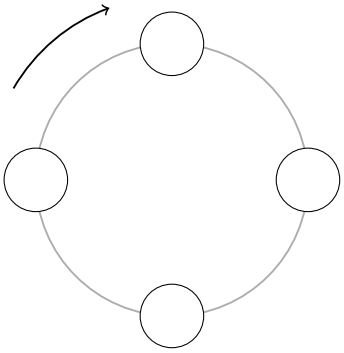


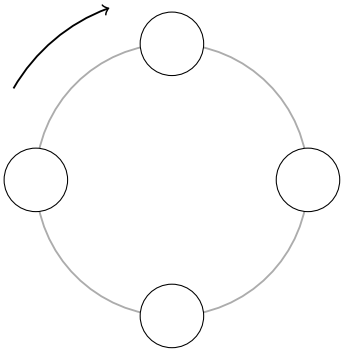


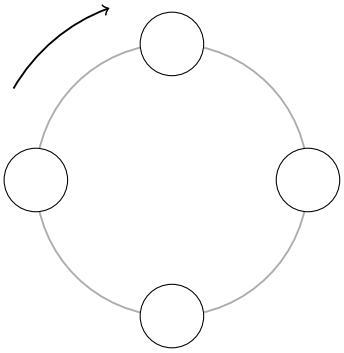


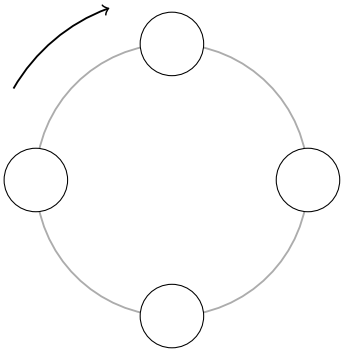


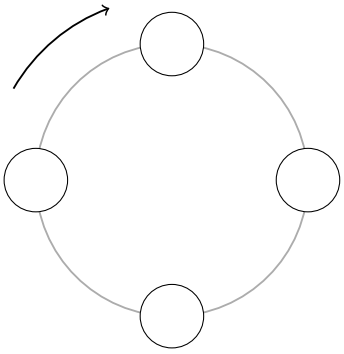
Problem 3: Make four-bead rings with exactly one, two, three or four different readouts. Which numbers are possible? Explain how you know.

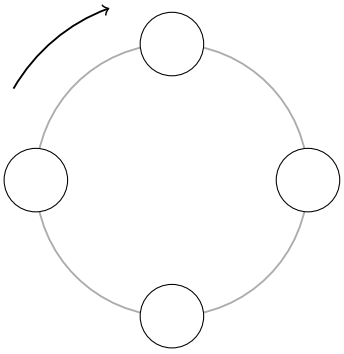




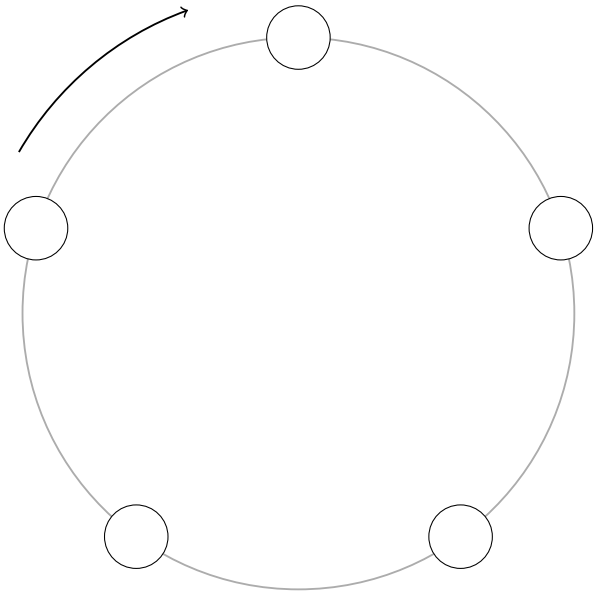


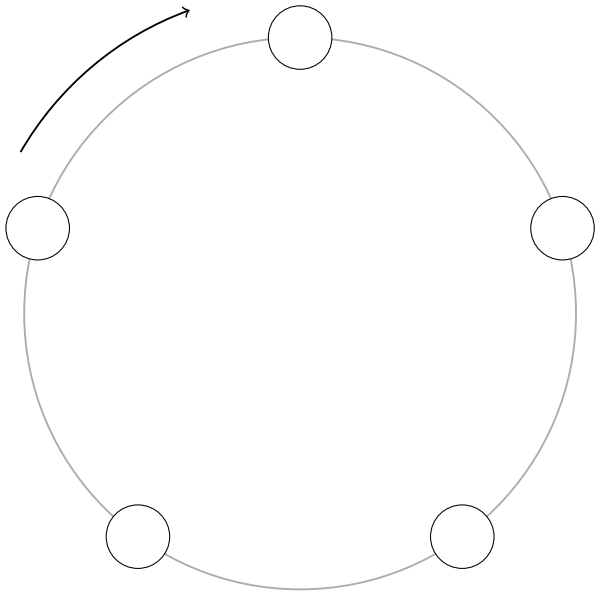




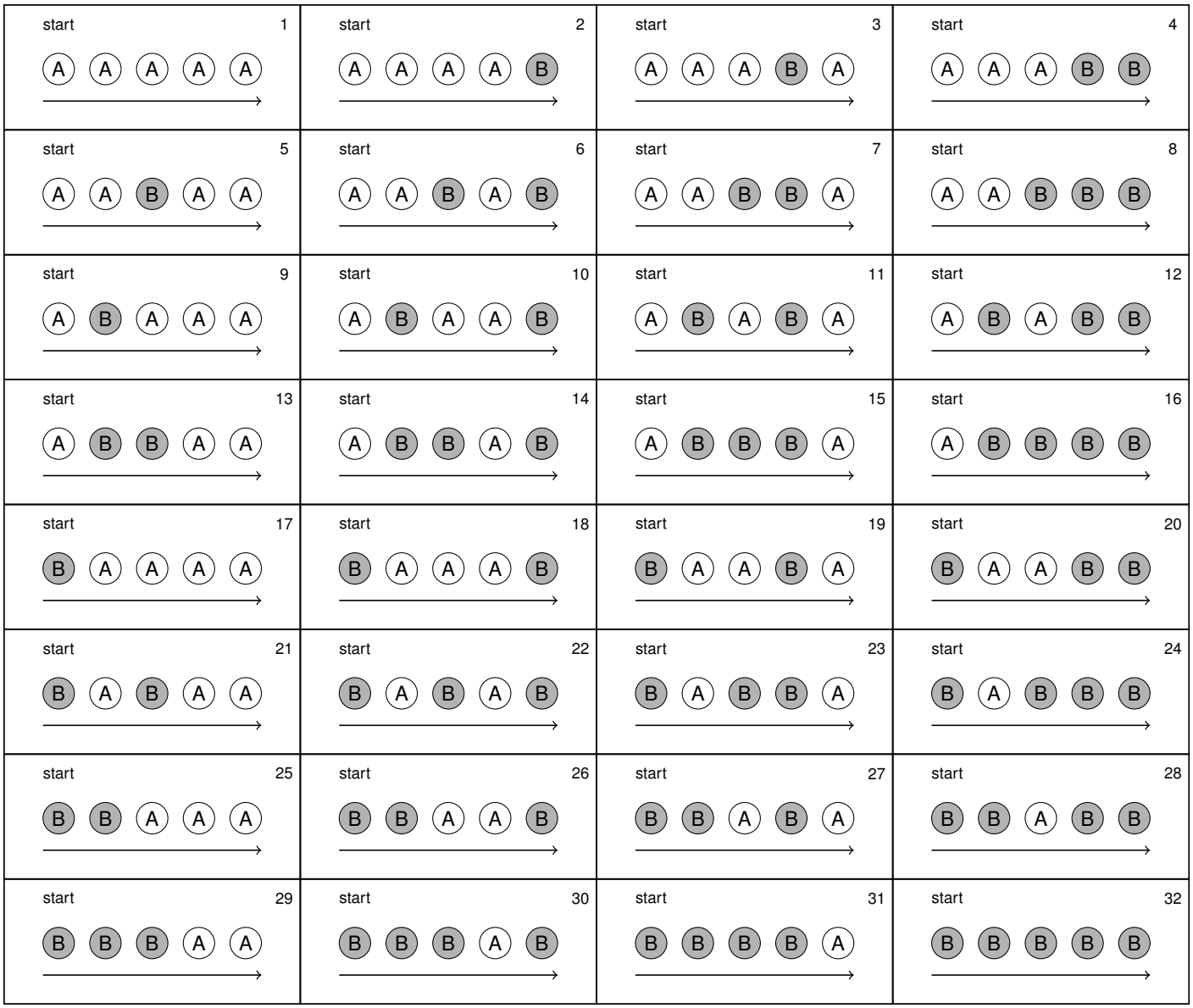


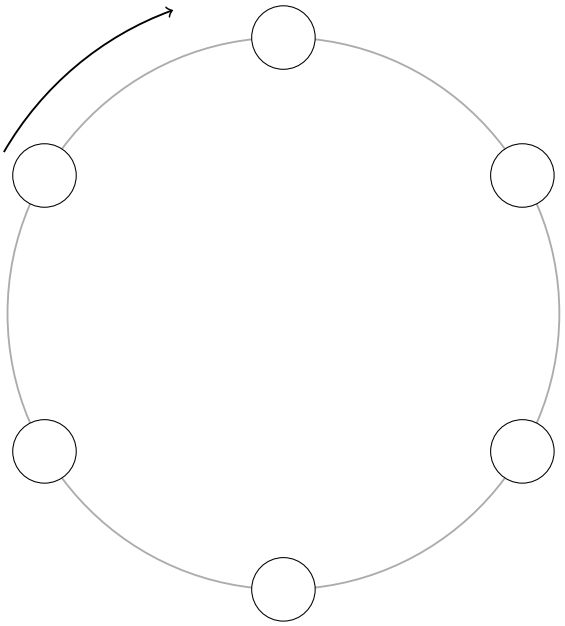
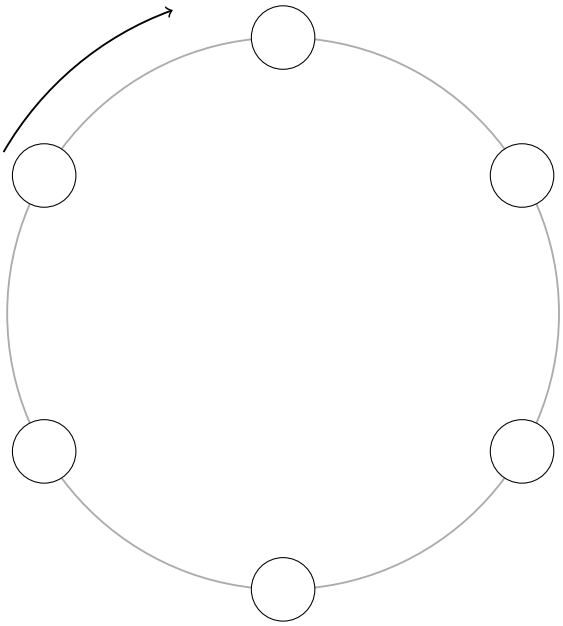
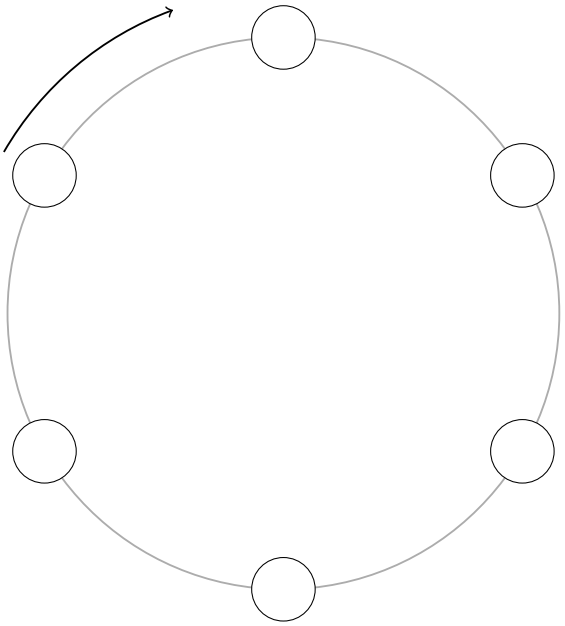
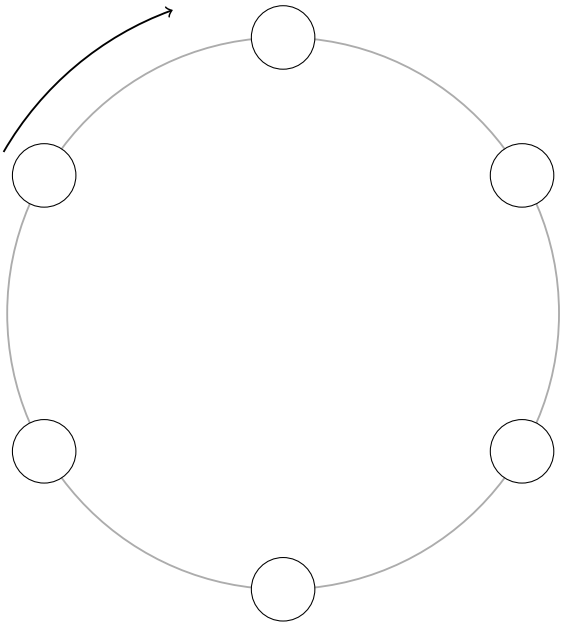
Problem 4: Can a five-bead ring using both A and B have fewer than five different readouts? Find one or explain why it cannot happen.





Problem 5: Group these five-bead cards by the rings they make. Find the size of each group and the total number of different rings.



Problem 6: Someone divides 32 by 5 to count the five-bead rings. What went wrong? Find a calculation that counts them correctly.

Problem 7: Use both A and B on a six-bead ring. Find rings with two, three and six different readouts.